

## GRADUATION PROJECT FUNDING PROPOSAL

# Design and Implementation of a Cooperative Autonomous Surface Vessel Swarm for Nile Surface Cleaning and Water-Hyacinth Handling

MSA University - Faculty of Engineering - Mechatronics Systems Engineering

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**REQUESTED FUNDING: EGP 40,000**

### EGYPT VISION 2030



### UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS



*Target invasive species: Pontederia crassipes (water hyacinth; syn. Eichhornia crassipes).  
Locally developed autonomous technology for cleaner, safer and more maintainable Egyptian waterways.*

# 1. Project information

## Arabic project title:

(Pontederia crassipes) تصميم وتنفيذ سرب تعاوني من المركبات السطحية ذاتية التشغيل لتنظيف سطح نهر النيل والتعامل مع ورد النيل

English project title: Design and Implementation of a Cooperative Autonomous Surface Vessel Swarm for Nile Surface Cleaning and Water-Hyacinth Handling

Department: Mechatronics Systems Engineering

Faculty: Faculty of Engineering

University: MSA University

Project team: Aiham Moustafa Awad (ID 234825) and Omar Mohamed Bakr (ID 235395)

Academic supervisors: Assoc. Prof. Dr. Amgad M. Bayoumy Aly and Prof. Dr. Mostafa Zaki Mohamed

Project sponsor, if formally confirmed before submission: \_\_\_\_\_

# 2. Project description in Arabic

يهدف المشروع إلى تطوير سرب تعاوني مكون من 3 إلى 4 مركبات سطحية ذاتية التشغيل لتنظيف سطح نهر النيل من المخلفات الطافية والتعامل تُصمَّم وتُصنَّع بهيكل فيبرجلاس جديد، ومركبة (Robot A) يتكوّن الأسطول من مركبة جديدة واحدة. (Pontederia crassipes) مع ورد النيل يتم استلامها وفحصها وقياس أداؤها وتحديثها، بالإضافة إلى مركبة أو مركبتين من منصات طلابية/مشروعات سابقة (Robot B) سابقة MSA يعاد توظيفها حسب حالتها الفنية. لا يشترط تطابق المركبات ميكانيكياً؛ وإنما يتم توحيد واجهات الملاحة والاتصال والحالة والمهام. تعتمد المنظومة اقتصادي ومناسب للدور على Linux أو منصة طيار آلي بحرية مناسبة، مع حاسب STM32 على تحكم منخفض المستوى وحماية أنية باستخدام وبوصلة مغناطيسية، ويستخدم الاتصال بين GNSS/IMU والرؤية والذكاء الاصطناعي. تستخدم الملاحة ROS 2 المركبات التي تحتاج لتبادل رسائل الحالة والمهام بصورة لامركزية دون الحاجة إلى حاسب شاطئي STM32 المتصلة مباشرة بالـ SX1262 LoRa المركبات وحدات مدرّب/مهياً باستخدام بيانات YOLO ونموذج RGB دائم؛ ويمكن استخدام حاسب محمول للمراقبة والضبط فقط. تعتمد الرؤية على كاميرا منخفض التكلفة على المركبات التي يثبت احتياجها إليه. لا يتضمن خط الأساس LiDAR للمدى القريب عند الرسو و ToF المنشورة، مع FloW أو التمويل المطلوب أي رادار. يشمل العمل منظومة جمع المخلفات، والتعامل الميكانيكي مع ورد النيل مع تفصيل الانقاط الكامل وتقليل التقنيات، مخصصة عند ثبوت PCB وقاطع/مغزٍ مساعد فقط إذا أثبتت الاختبارات الحاجة إليه، ومنظومة رسو وخدمة وتفرغ. كما يتضمن التصميم لوحات أنها تقلل الأسلاك وتحسن الاعتمادية. فلسفة الشراء هي استخدام أقل مكون تكلفة يحقق متطلبات الأداء والسلامة والاعتمادية المقاسة، مع إعادة استخدام أي مكونات قديمة سليمة بدلاً من شراء مكونات مرتفعة السعر لمجرد أنها الأفضل في فئتها.

# 3. Project description in English

This project will develop a cooperative physical swarm of approximately three to four autonomous surface vessels for floating-waste collection and water-hyacinth (Pontederia crassipes; syn. Eichhornia crassipes) handling in Nile environments. The fleet baseline consists of one new fiberglass vessel designed and fabricated from scratch (Robot A), one continued and upgraded MSA cleaning vessel from the previous project (Robot B), and one or two technically viable legacy boats repurposed as Robot C and optional Robot D. The vessels are intentionally heterogeneous: they share minimum navigation, communication, health-state and task interfaces rather than identical hardware. Each participating vessel uses deterministic low-level control and safety on an STM32-based controller or supported marine-autopilot hardware, while a role-appropriate Linux SBC is fitted only where ROS 2, vision, mission logic or AI require it. Navigation uses GNSS, IMU and an external magnetic compass. SX1262 LoRa transceivers connected directly to the STM32 provide low-bandwidth peer status/task exchange for decentralized cooperation, so no permanent shore computer or master vessel is required; a laptop may be used for supervision, configuration and data offload. Floating-waste perception uses RGB vision and a YOLO model based on the published FloW image dataset. ToF is used for close-range docking, while low-cost 2D LiDAR may be concentrated on selected vessels only if tests and budget justify it. No radar is included in the baseline system or requested budget. Robot A includes a new fiberglass hull, and the project may use custom PCB prototypes for STM32/SX1262 interfaces, protected sensor I/O and power/health monitoring. Physical payloads include floating-waste collection, water-hyacinth pickup/feed with conditional cutting only where tests

justify it, anti-entanglement measures, and a docking/unloading service cycle. Procurement is performance-based and reuse-first: the project will buy the lowest-cost locally serviceable component that demonstrably meets measured mission, safety and reliability thresholds rather than specifying best-in-class brands.

## 4. General objective and problem addressed

The general objective is to design, implement and experimentally evaluate a cooperative physical swarm of approximately three to four autonomous surface vessels that can remove floating waste and handle water hyacinth (*Pontederia crassipes*) in representative Nile conditions. The swarm will combine one new fiberglass vessel, one continued inherited vessel and one or two repurposed legacy platforms under a common decentralized communication, task and safety architecture without requiring a permanent shore computer.

The problem has four connected parts. First, floating solid waste and dense *Pontederia crassipes* mats require repeated physical collection, separation, retention and servicing. Second, the Nile environment introduces current, wind, fouling, corrosion, entanglement, waterproofing and docking requirements. Third, a practical multi-vessel system must remain safe and useful after communication loss or withdrawal of one vessel rather than depending on a single master. Fourth, the project must achieve these functions under a limited graduation-project budget; therefore propulsion, computing, sensing, batteries and fabricated mechanisms must be selected from measured performance requirements rather than from premium-brand specifications.

The project therefore uses the inherited robot as a measured baseline, builds only one completely new fiberglass vessel, and extracts additional value from one or two older boats that can be repurposed. Common interfaces will be standardized while expensive hardware remains role-specific. A sensor-rich vessel may carry stronger compute or LiDAR, while a simpler collector/scout may use cheaper or reused hardware if it meets the same mission interface and safety requirements. This approach directs the requested funding toward genuine capability gaps, custom mechanical/electrical integration and field reliability while avoiding unnecessary fleet-wide duplication.

## 5. Work methodology, objective indicators, outputs and expected returns

The project will follow an evidence-before-freeze, performance-based engineering process. Robot B will first be documented, weighed and tested as received, while Robot C and optional Robot D are inventoried for usable hull, propulsion, battery, sensing and payload capacity. Robot A will then be designed as a new fiberglass vessel using hydrostatics, stability, resistance, thrust, current, power and payload calculations. Propulsion candidates will be bench-tested for thrust, current and temperature before purchase is repeated across the fleet. Water-hyacinth mechanisms will be bench-tested for feedability, cutting load, jam behavior, retained biomass and fragment escape before installation. The low-level control architecture (supported marine autopilot versus custom STM32) will be frozen before harness/PCB duplication. Role-appropriate Linux SBCs will be accepted by workload, not model name, using ROS 2 plus FloW/YOLO inference where required. GNSS/IMU/compass navigation, SX1262 LoRa peer communications, ToF docking and selected-vessel LiDAR will be verified independently. Single-vessel autonomy and safety will be accepted before decentralized CBBA-style task allocation, cutter-collector cooperation and autonomous docking/service trials are enabled.

MATLAB/Simulink/MSS, ANSYS Fluent and VRX/Gazebo will provide physics and repeatable current/wind simulations, but physical tests remain the acceptance authority. Representative Nile trials will occur only when permissions, recovery arrangements and safety support are in place.

Indicators used to verify the objectives will include:

- Navigation and docking: GNSS/IMU/compass cross-track and heading error, waypoint/coverage completion, repeatability, close-range docking error and autonomous docking success.
- Propulsion and energy: measured thrust, current and temperature; initial design target of approximately 0.8-1.0 m/s fully loaded in calm/low-current water, positive upstream progress against the representative design

current, at least 15% usable thrust reserve, and at least 60 minutes representative mixed-duty operation with a 25% protected return reserve.

- Waste collection: standardized mixed-debris capture target of at least 80%, retention/backflow, clogging, wet load, power consumption, bin-full detection and unloading time.
- Water hyacinth (*Pontederia crassipes*): intact-pickup/cutting success as applicable, kg/h and Wh/kg, jam and entanglement frequency, retained versus escaped biomass, peak current and deterministic jam recovery.
- Communications and fleet behavior: SX1262 LoRa heartbeat/status delivery target of at least 95% across the operational test area, task ownership, CBBA/consensus convergence, reassignment time, unfinished-work recovery and mission completion after a vessel withdrawal or partition.
- Safety, service and reliability: leak events, manual override, safe Hold/RTL behavior, cutter/thruster protection, docking target of at least 8 successful representative autonomous docks in 10 trials, safe unloading and recorded failure families.

Expected returns include a reusable local autonomous-marine-robotics platform, documented Nile-oriented propulsion/current/stability data, practical *Pontederia crassipes* handling and anti-entanglement knowledge, reusable custom electronics/PCB designs, and a physical 3-4 vessel cooperative swarm. The mixed new-build, continued and repurposed fleet will demonstrate a value-engineered approach in which locally available legacy platforms and affordable components are accepted by measured capability rather than by brand or identical hardware configuration.

## 6. Main outputs and potential final beneficiaries

- Robot B: recovered and upgraded inherited MSA cleaning vessel with an as-received baseline dataset and measured propulsion/collection performance.
- Robot A: one newly engineered autonomous cleaning vessel with a fiberglass hull designed and fabricated from scratch using verified buoyancy, stability, propulsion, power, collection and service requirements.
- Robot C and optional Robot D: one or two repurposed legacy boats upgraded, where technically viable, to the common decentralized communication, health-state, task and safety interfaces.
- A validated floating-waste collector and *Pontederia crassipes* handling subsystem with measured capture, jam, entanglement, power, retained-biomass and fragment performance; cutting remains conditional on test evidence.
- A dual-layer navigation/control architecture using GNSS/IMU/compass, deterministic STM32 or supported marine-autopilot safety/control, role-appropriate onboard Linux computing for ROS 2/perception, and synchronized logging/diagnostics.
- Decentralized peer-to-peer fleet coordination using STM32-connected SX1262 LoRa, shared task state and capability-aware allocation without a permanent master vessel or mandatory shore computer.
- A practical docking, staging, loading/unloading and optional charging concept with measured service-cycle performance using camera marker alignment, ToF and hard service interlocks.
- Custom PCB prototype(s) where justified for STM32/SX1262 interfaces, protected sensor/power-health I/O and maintainable wiring, together with a repeatable experimental dataset, source-controlled models/code and complete engineering documentation.

## 7. Required budget and itemized justification

**Total requested funding: EGP 40,000.**

The requested budget is intentionally value-engineered and all allocations are fleet-wide procurement ceilings. It does not seek best-in-class hardware; it funds the lowest-cost locally serviceable components that can demonstrate the required thrust, endurance, compute, navigation, sensing, collection, docking and safety performance. Serviceable legacy propulsion, batteries, cameras, computers and sensors will be reused after testing. Robot A receives a dedicated fiberglass/structural allocation, custom PCB prototypes and waterproof

integration are budgeted explicitly, and docking/unloading is treated as a distinct subsystem. No radar procurement is included. Final part selection will follow measured acceptance criteria and may differ between vessel roles.

| Budget area                                                            | Requested allocation         | What the allocation covers                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>7.1 Propulsion, steering and motor-control hardware</b>             | EGP 9,000<br>(22.5%)         | Fleet-wide selective replacement of missing or inadequate marine propulsors/thrusters, compatible ESCs or motor drivers, mounts, guards and anti-fouling protection. Serviceable legacy propulsion will be reused. Final selection is based on measured thrust, current, temperature, loaded-vessel speed and steering authority rather than brand.                                                                                                   |
| <b>7.2 Battery, charging and protected power distribution</b>          | EGP 5,000<br>(12.5%)         | Fleet-wide ceiling after auditing reusable batteries and chargers. New traction batteries are purchased only where existing packs fail the required duty cycle, discharge-current or health tests; the allocation also covers BMS/protection, fuses, isolation, DC-DC conversion, high-current wiring, charging hardware and whole-pack current/temperature monitoring. No fixed 50 Ah pack is assumed for every vessel.                              |
| <b>7.3 Computing, navigation, sensing and LoRa communications</b>      | EGP 6,000<br>(15%)           | Role-appropriate Linux SBCs only where needed, STM32 electronics, GNSS/IMU/compass, SX1262 LoRa modules, RGB camera, ToF and required interface/driver electronics. No Raspberry Pi 5 is mandatory; reused or lower-cost compute is acceptable if it meets the workload. Low-cost LiDAR is conditional on selected-vessel need. No radar is funded.                                                                                                   |
| <b>7.4 Robot A fiberglass hull and structural fabrication</b>          | EGP 5,500<br>(13.75%)        | Fiberglass cloth/mat, resin, gelcoat/finish, reinforcement or core materials, inserts, bonding/sealing materials, local mold/fabrication work and structural mounts for the one new-build Robot A hull, sized from verified displacement, freeboard, stability, payload and propulsion requirements.                                                                                                                                                  |
| <b>7.5 Collector and water-hyacinth cutter/feed mechanisms</b>         | EGP 4,500<br>(11.25%)        | Waste-collection guides, net/belt/bin hardware and candidate Pontederia pickup/feed or cutting hardware where bench tests justify it: motor/gearmotor, bidirectional driver, reduction/transmission, shafts, bearings, replaceable cutting elements, guards and current sensing. Intact pickup remains preferred where practical.                                                                                                                     |
| <b>7.6 Custom PCB prototypes and waterproof electrical integration</b> | EGP 4,000<br>(10%)           | Prototype PCB fabrication/assembly for the common STM32/SX1262 and protected sensor/power-health interface where justified, including allowance for revision, plus waterproof connectors and cable glands, sealed electronics enclosures, leak sensing, high-current connectors, corrosion-resistant hardware and wiring protection. High-current thruster ESC power stages remain replaceable commercial modules unless testing justifies otherwise. |
| <b>7.7 Docking and unloading prototype</b>                             | EGP 2,500<br>(6.25%)         | Dock guides/fenders, marker mounting, ToF/contact or limit sensing, mechanical stop/latch hardware and a practical basket/gate/conveyor unloading prototype. Optional charging contacts are included only if the basic dock-and-unload cycle is proven first.                                                                                                                                                                                         |
| <b>7.8 Testing, spares, transport and contingency</b>                  | EGP 3,500<br>(8.75%)         | Test fixtures and consumables, critical replacement parts, local machining or small fabrication corrections, fasteners, transport and contingency for repeated bench, controlled-water and representative field trials. This reserve prevents a minor failed component from stopping the validation program.                                                                                                                                          |
| <b>TOTAL REQUESTED FUNDING</b>                                         | <b>EGP 40,000<br/>(100%)</b> | <b>Funding is performance-based, reuse-first and fleet-wide. Allocations are procurement ceilings rather than mandatory purchases: working legacy hardware is retained, expensive components are not duplicated unless tests prove the need, and every new part is accepted against measured mission, safety and reliability thresholds.</b>                                                                                                          |

### Performance-based procurement and acceptance criteria

| Subsystem / procurement rule | Initial acceptance target | Rationale |
|------------------------------|---------------------------|-----------|
|------------------------------|---------------------------|-----------|

| Subsystem / procurement rule       | Initial acceptance target                                                                                                                                                                                                                | Rationale                                                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propulsion and steering            | Loaded cruise approximately 0.8-1.0 m/s in calm/low-current water; positive upstream ground progress against the representative design current; at least 15% usable thrust reserve; no thermal/current-limit failure in sustained tests. | Buy/reuse the lowest-cost marine propulsion that passes measured thrust, steering, current and temperature tests; no premium thruster requirement.     |
| Onboard computing                  | ROS 2 + selected FloW/YOLO inference at at least 5 useful FPS while mission logic, logging and communications are active, with no sustained thermal throttling.                                                                          | No mandatory Raspberry Pi 5. Reuse or buy a role-appropriate Linux SBC; stronger compute is fitted only to boats that actually need onboard vision/AI. |
| Battery and protected power        | At least 60 minutes representative mixed-duty operation plus 25% protected return reserve; BMS, fuse/isolation, current and temperature monitoring verified.                                                                             | Capacity is selected from measured duty cycle rather than buying a fixed 50 Ah pack for every vessel.                                                  |
| Navigation, ranging and fleet link | GNSS + IMU + compass baseline; RGB camera and ToF; selected-vessel low-cost LiDAR only if justified; SX1262/STM32 peer link with at least 95% heartbeat/status delivery in the operational test area. No radar.                          | Standardize the mission interface, not identical expensive sensors. Close-range docking and robust status exchange receive priority.                   |

*These are initial engineering acceptance targets for procurement and prototype decisions. Final thresholds will be refined after Robot B baseline measurements and representative-current testing. Equivalent local or reused parts may be substituted whenever they meet the same measured requirement; brand/model is not fixed.*

## 8. Potential conversion into a company

Yes, conditionally. A company would only be justified after the project demonstrates repeatable cleaning and hyacinth-handling performance, safe decentralized autonomous operation, practical docking/servicing and unloading, low maintenance burden, local fiberglass/mechanical manufacturability, maintainable custom electronics, and acceptable lifecycle cost. The most realistic path is a modular Egyptian waterway-service platform in which propulsion, compute and sensors are selected by required capability and serviceability rather than premium-brand specification.

## 9. Two-year vision and desired partnerships

Within two years, the target is to progress from the 3-4 vessel graduation-project swarm into a field-validated pilot architecture with improved payload, service/docking, anti-entanglement, local fiberglass fabrication, custom electronics and maintainability. The first year will focus on completing Robot A, upgrading Robot B, repurposing Robot C and optional Robot D where feasible, validating *Pontederia crassipes* handling, propulsion/current response, decentralized communications/task reallocation and physical service cycles. The second year would focus on pilot-scale refinement, repeatable maintenance, cost reduction, local sourcing and deciding whether startup formation, licensing or technology transfer is justified by field evidence.

Desired future partnerships include the National Water Research Center and relevant Ministry of Water Resources and Irrigation entities for requirements and representative testing, local marine/fabrication workshops

for manufacturability, Egyptian electronics and propulsion suppliers for maintainable sourcing, and environmental-cleaning operators for deployment feedback. No partnership is represented as confirmed unless formally documented.

## 10. Supervisor question: previous supervision of a similar project

Previous similar project supervised: “Autonomous Water Surface Cleaning Robot,” MSA University, academic year 2025/2026. The present project continues that technical line while establishing a new measured baseline, adding one new fiberglass vessel and one or two repurposed legacy vessels to form a 3-4 vessel physical swarm, introducing value-engineered role-specific hardware, decentralized peer coordination, Pontederia crassipes handling, docking/service, current/disturbance testing and cooperative fleet operation.

Supervisor confirmation/name: \_\_\_\_\_